MODULE 03 · DOMAIN 1 - SOLUTION DESIGN AND ARCHITECTURE

Patterns: Augmented LLM, Workflows, Agents

The building block, five workflow patterns, and the point at which an agent earns its cost.

DOMAIN 1 of 7 EXAM WEIGHT 17% MODULE 3 of 38

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A workflow follows the path you wrote. An agent chooses its own. You pay for that freedom in **cost, latency and predictability**.

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The augmented LLM: the building block

Retrieval
Your documents
Fetched per request

Tools
Look things up
Take actions

Claude
Your prompt
Your examples
One model call, extended

Memory
State across turns
Notes kept outside

THE TAKEAWAY

Every pattern in this module is built from augmented LLM calls, arranged differently.

Workflow or agent?

WORKFLOW

Code decides the path

- Steps known in advance
- Predictable cost and latency
- Each step testable on its own
- Easy to audit

AGENT

The model decides the path

- Steps cannot be predicted
- Turns and cost vary per task
- Needs limits and checkpoints
- Test in a sandbox first

THE TIEBREAKER

If you can draw the flowchart before the request arrives, build a workflow.

Five workflow patterns, five signals

PATTERN	CHOOSE IT WHEN
Prompt chaining	Fixed subtasks, in a set order
Routing	Inputs fall into distinct types
Parallelisation	Independent parts, or several opinions
Orchestrator-workers	Subtasks depend on the input
Evaluator-optimiser	Clear criteria, and critique improves drafts

> ON THE EXAM
>
> Learn the signal, not just the name. The exam describes the situation, never the pattern.

A Mereside complaint, step by step

01
Classify
Product and complaint type.

02
Extract
Dates, amounts, what went wrong.

03
Gate
Code checks the fields are complete.

04
Draft
A response in house style.

05
Gate
Mandatory wording present?

06
Approve
A complaints handler signs off.

ON THE EXAM

Gates are programmatic checks between steps. They catch errors before they compound.

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Classify first, then send each type to its best handler

- **Separate prompts and tools per type**, so tuning one does not break another.
- **Separate tiers per type:** easy traffic to a light model, hard cases to a capable one.
- **The classifier must be good:** a wrong route gives a confident wrong answer.
- **Example:** general questions, refunds and technical faults, each to its own flow.

> ON THE EXAM
>
> Routing is a quality pattern and a cost pattern. Send the easy majority to the light tier.

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Sectioning or voting?

SECTIONING

Different pieces at once

- Split independent parts
- Each call does one part
- Combine the results
- Example: screen and answer together

VOTING

The same piece, several times

- Run the same task repeatedly
- Or several prompts, one question
- Combine with a threshold
- Example: code vulnerability review

THE TIEBREAKER

Different pieces at once is sectioning, for speed. The same piece several times is voting, for confidence.

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Orchestrator-workers or parallelisation?

PARALLELISATION

You define the subtasks

- Fixed list, written in code
- Same split for every input
- Example: five standard sections

ORCHESTRATOR-WORKERS

The model defines them

- A lead call plans at run time
- Workers do the pieces
- The lead combines results
- Example: a change across many files

THE TIEBREAKER

If the subtasks are known before the input arrives, it is parallelisation. If not, orchestrator-workers.

One call drafts, another critiques, until it passes

- **Use it when** criteria are clear and critique measurably improves the result.
- **The evaluator checks against explicit criteria** and returns specific feedback.
- **Always cap the loop**, then hand unresolved cases to a person.
- **Examples:** literary translation; letters that must meet set rules.

> ON THE EXAM
>
> No clear criteria, no evaluator-optimiser. The loop needs something definite to aim at.

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When an agent earns its cost

- **Open-ended:** the number of steps cannot be predicted.
- **Valuable enough** to pay for variable turns, tokens and time.
- **Success is checkable** from the environment: tests pass, data found.
- **Always with limits:** a turn cap, scoped tools, a sandbox, human checkpoints.

> ON THE EXAM
>
> Agents are simple to build and hard to bound. The bounds are the architecture.

Reading the signal

WHEN THE BRIEF SAYS	CHOOSE
Same steps every time, audit each one	Prompt chaining
Several kinds of request, different handling	Routing
Independent parts, results needed fast	Sectioning
Missing a problem is costly; want agreement	Voting
What needs doing depends on the request	Orchestrator-workers
Clear rubric; drafts keep missing it	Evaluator-optimiser
Unknown steps, explore and adapt	Agent, with limits

ON THE EXAM

Several signals at once? Patterns combine. A router can send one type to a chain and another to an agent.

> Mereside handles about 3,000 complaints a month. Each goes through the same steps: classify it, extract the facts, draft a response, check it against complaint-handling rules, and pass it to a handler.
>
> The model risk committee wants a predictable cost per complaint and a record of what happened at every step.

Which pattern fits best?

A An autonomous agent with tools for the complaints case system

B A prompt chain in code, with programmatic checks between steps

C A team of agents: a planner, a researcher and a writer

D One large prompt that performs every step in a single call

◆ PAUSE AND DECIDE

Which pattern fits best?

A An autonomous agent with tools for the complaints case system

The steps are known. An agent adds variable cost and a harder audit.

B A prompt chain in code, with programmatic checks between steps

Fixed steps, predictable cost, every step recorded and gated.

C A team of agents: a planner, a researcher and a writer

There is nothing to plan. Multi-agent cost for a fixed sequence.

D One large prompt that performs every step in a single call

Cheaper, but no intermediate record and no gate before the draft.

THE SIGNAL

Same steps every time, audit at each one: prompt chaining with gates.

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> Tallyhawk's support assistant answers about 55,000 conversations a month. Roughly 70% are simple how-to questions, 20% are billing queries needing account tools, and 10% are complex integration faults.
>
> Every conversation currently goes to the most capable tier with one general prompt. Costs are rising and billing answers are inconsistent.

What should Corvane change?

A. Keep the capable tier for everything, and lengthen the general prompt

B. Move every conversation to the lightest tier to cut cost

C. Classify each conversation first, then route each type to its own prompt, tools and tier

D. Add an evaluator that critiques and redrafts every reply before it is sent

◆ PAUSE AND DECIDE

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What should Corvane change?

A Keep the capable tier for everything, and lengthen the general prompt

Costs stay high, and one prompt still serves three different jobs.

B Move every conversation to the lightest tier to cut cost

Cheap, but the hard integration faults lose the reasoning they need.

C Classify each conversation first, then route each type to its own prompt, tools and tier

Distinct types get their own handling; the easy majority goes to the light tier.

D Add an evaluator that critiques and redrafts every reply before it is sent

Adds a call to every reply. Raises cost, and fixes neither cause.

THE SIGNAL

Distinct types with different needs, and cost pressure: routing.

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A commercial insurer reviews broker submissions in its portal. Every submission has the same five independent sections, and each is checked against its own underwriting guidelines.

Running the five checks one after another takes over a minute, and brokers are waiting on screen.

Which design best cuts the wait?

A Run the five section checks as separate calls in parallel, then combine the results

B One call that reviews all five sections against all the guidelines at once

C An orchestrator that decides at run time which sections each submission needs

D An evaluator-optimiser loop over the whole submission

◆ PAUSE AND DECIDE

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Which design best cuts the wait?

A Run the five section checks as separate calls in parallel, then combine the results

Independent, fixed sections: sectioning cuts the wait to the slowest single check.

B One call that reviews all five sections against all the guidelines at once

One overloaded call. Each focused check is more reliable.

C An orchestrator that decides at run time which sections each submission needs

The sections are always the same five. Nothing to decide at run time.

D An evaluator-optimiser loop over the whole submission

Adds rounds of calls, so the wait gets longer, not shorter.

THE SIGNAL

Fixed, independent parts and a latency problem: sectioning.

> Tallyhawk's security team wants Claude to screen every pull request for injection flaws, leaked secrets and unsafe permission changes.
>
> A missed problem is far more costly than a false alarm, and engineers are happy to look again at anything flagged.

Which pattern best fits the screening?

A One review call at the highest effort setting

B A chain that summarises the change, then reviews the summary

C Route each pull request by programming language to a specialist prompt

D Several independent reviews with different focuses, flagging if any one finds a problem

◆ PAUSE AND DECIDE

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Which pattern best fits the screening?

A One review call at the highest effort setting

Better than a quick look, but one opinion is one chance to miss.

B A chain that summarises the change, then reviews the summary

Summaries drop detail, and the flaw may be in the detail.

C Route each pull request by programming language to a specialist prompt

Language is not the axis that matters. Each route is still one opinion.

D Several independent reviews with different focuses, flagging if any one finds a problem

Voting with a low threshold: misses are costly, false alarms are cheap.

THE SIGNAL

Misses are costly and false alarms are cheap: voting, with a low threshold to flag.

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> Tallyhawk wants an engineering assistant that implements small feature requests across its monorepo and roughly 30 services.
>
> Which files and services a request touches is only clear once the request has been analysed. Some touch one file; others touch a dozen across four services.

Which pattern fits best?

A. A prompt chain with the same five fixed steps for every request

B. An orchestrator that plans the subtasks for each request, delegates them, and integrates the results

C. Route each request to the prompt for the service it mentions first

D. Run one call per service, in parallel, for every request

◆ PAUSE AND DECIDE

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Which pattern fits best?

A. A prompt chain with the same five fixed steps for every request

Fixed steps cannot fit work whose shape differs per request.

B. An orchestrator that plans the subtasks for each request, delegates them, and integrates the results

Subtasks depend on the input: exactly what orchestrator-workers is for.

C. Route each request to the prompt for the service it mentions first

Many requests span several services. One route misses the rest.

D. Run one call per service, in parallel, for every request

Thirty calls for a one-file change. The split ignores the input.

THE SIGNAL

Subtasks only knowable after reading the input: orchestrator-workers.

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Mereside's mortgage decline letters must meet a written checklist: plain English, the mandatory regulatory wording, the main reason for the decision, and the customer's next steps.

First drafts usually meet most of the checklist but often miss one item. An underwriter approves every letter.

Which design best raises the share of letters that pass first time?

A Raise the effort setting on the single drafting call

B Add twenty example letters to the drafting prompt

C A second call checks each draft against the checklist and feeds back; redraft until it passes or a limit is reached

D An autonomous agent that drafts, checks and sends letters without the underwriter

◆ PAUSE AND DECIDE

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Which design best raises the share of letters that pass first time?

A Raise the effort setting on the single drafting call

May help a little, but nothing checks the draft against the list.

B Add twenty example letters to the drafting prompt

Examples steer style; they do not verify each item was met.

C A second call checks each draft against the checklist and feeds back; redraft until it passes or a limit is reached

Clear criteria, and critique fixes the missing item: evaluator-optimiser, capped.

D An autonomous agent that drafts, checks and sends letters without the underwriter

Removes the required human decision on a high-risk outcome.

THE SIGNAL

A written checklist and drafts that miss one item: evaluator-optimiser, with a loop limit.

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> Tallyhawk customers connect it to many third-party accounting systems. When a connection fails, the cause could be credentials, field mappings, rate limits or a supplier outage, and each case needs a different trail of investigation.
>
> Investigation can run in a sandbox with read-only access to logs and settings. A support engineer must approve any fix.

Which design fits the diagnosis?

A. A fixed four-step chain: read logs, read settings, diagnose, propose a fix

B. One call given every log and setting the customer has

C. Route each failure to a prompt for its accounting system

D. An agent with read-only diagnostic tools, a turn limit, and hand-off of its proposed fix to an engineer

◆ PAUSE AND DECIDE

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Which design fits the diagnosis?

A. A fixed four-step chain: read logs, read settings, diagnose, propose a fix

Each case follows a different trail. A fixed path cannot follow the evidence.

B. One call given every log and setting the customer has

Floods the context, and cannot go and check what the first look suggests.

C. Route each failure to a prompt for its accounting system

Routes on the wrong axis. The cause, not the supplier, decides the trail.

D. An agent with read-only diagnostic tools, a turn limit, and hand-off of its proposed fix to an engineer

Open-ended investigation, bounded by read-only tools, a cap and a human.

THE SIGNAL

Unpredictable steps that follow the evidence: an agent, bounded by tools, limits and a human.

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> Quillon is designing prior-authorisation letters. The steps are known: gather the relevant record sections, map them to the insurer's criteria, draft, and check. Compliance wants the same steps in the same order for every letter, and finance wants a predictable cost per letter.
>
> Records vary in length. The team is keen to try an agent framework. Every letter needs a clinician's signature.

Which two details most favour a workflow over an agent? Choose two.

- A Compliance wants the same steps in the same order for every letter
- B Records vary in length from case to case
- C The team is keen to try an agent framework
- D Every letter needs a clinician's signature
- E Finance wants a predictable cost per letter

◆ PAUSE AND DECIDE

Which two details most favour a workflow over an agent? Choose two.

A Compliance wants the same steps in the same order for every letter

A fixed, auditable path is the definition of a workflow.

B Records vary in length from case to case

Length changes input size, not the steps. Either pattern copes.

C The team is keen to try an agent framework

A preference. It decides nothing about the pattern.

D Every letter needs a clinician's signature

Human sign-off belongs in both patterns, so it does not separate them.

E Finance wants a predictable cost per letter

Workflows give predictable tokens per run; agents vary.

THE SIGNAL

Fixed order and predictable cost point to a workflow. A human gate fits either pattern.

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In five lines

- **Augmented LLM**: one call with retrieval, tools and memory. Often enough.
- **Workflow** when you can draw the flowchart; **agent** when you cannot.
- **Chain** fixed steps; **route** distinct types; **section** for speed; **vote** for confidence.
- **Orchestrator-workers** when subtasks depend on the input; **evaluator-optimiser** with clear criteria.
- **Agents need bounds:** turn caps, scoped tools, sandboxes, human checkpoints.

> COMING UP
>
> Next: multi-agent systems. When a team of agents is worth fifteen times the tokens, and when it is not.